

Click, Trust, Repeat: How Social Media Turns Humans into Cybersecurity Loopholes

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WRTG 2010 Introduction to Writing

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10 professional academic sources

I. Introduction (150–250 words)

- **Hook:** Start with a striking statistic or example (e.g., “Over 80% of cybersecurity breaches involve human error, and many originate on social media platforms.”)
- **Context:** Explain how social media and cybersecurity intersect — people share, connect, and trust strangers online.
- **Problem:** Highlight how trust-based interactions can be exploited by hackers or scammers.
- **Thesis:** Present your central claim (from above).
- **Preview:** Briefly outline your key points (psychology, manipulation, data exposure, case studies, and solutions).

II. The Human Element in Cybersecurity (250–350 words)

- Explain why **humans are considered the weakest link** in security systems.
- Discuss cognitive biases (curiosity, trust, overconfidence).
- Include research on **human error and cyber risk**.
- Example source: *Journal of Cybersecurity* (Oxford Academic) – “Human Factors in Cybersecurity.”

Humans are the weakest link in the security of digital world. In general, cybersecurity is thought to be only related in technology and newtorking. However, its vision extends beyond these days due to recent optimized attacks held among social media users. According to the Journal of Cybersecurity, human behavior accounts for the vast majority of successful cyberattacks, simply because people are easier to get manipulated than a secure network with well-configured systems (Chaudhary et al., 2022). A common practice in a cyberattack is the human interaction known as social engineering. Social engineering is often accomplished by using characteristics of a person's psychology, such as curiosity, fear, and trust behavior. This leads an individual to to provide sensitive information or click a link that launches malware. In this case, the social engineer "hacks" the individual rather than the device and is able to exploit even the most sophisticated security systems. For example, an Instagram or Facebook message reads as an interaction from a trusted friend or brand and persuades the user to act without regard to their critical thinking process. With the increased use of digital conversations, individuals underestimate how many personal details they expose online, such as location tags and relationship information, all information that cyber criminals need to enrich their schemes. This means that a complete thought about cybersecurity in the future will involve more than just firewalls and encryption, it will involve understanding the physiological threats of human – most vulnerable bug for exploitation in social media. NEED EDIT

<https://academic.oup.com/cybersecurity/article/8/1/tyac006/6590603>

<https://www.proquest.com/openview/2661364c569d9dde0f31d87716716c96/1?pq-origsite=gscholar&cbl=18750>

III. Social Media and the Illusion of Trust (250–350 words)

- Analyze how social media fosters **false trust** — users feel safe sharing with “friends” or influencers.
- Show how attackers exploit this trust to steal information or manipulate people.
- Example: Romance scams, fake charity links, or “verified account” impersonations.
- Use studies on **online trust and parasocial relationships**.

The range of online social networks is broad when it’s dealing with false trust. Social media platforms are intentionally designed to make users feel safe and connected, encouraging them to share personal information with “friends,” “followers,” or familiar faces. This sense of belongingness creates an illusion of safety, where individuals do not realize that they are simply replicating trust behaviors that they would normally engage in face-to-face. Ershova (2024) explains that irrational trust in online networks not only alters an individual’s judgment, but also makes them psychologically defenseless, creating a false sense of security that generation of cyber criminals recognize and exploit. Attackers exploit this psychological defenselessness and false sense of security through manipulation strategies like downplaying outcomes like romance scams, fake charity accounts, and false identities. Khukhunaishvili (2024) adds that romance scams, as one of the most brutal violations of cyberspace scams rely on manipulating emotions over a long time before convincing the victim to financially and psychologically exploit them. Scammers often use social media and dating platforms to connect with targets due to the amount of personal experiences and emotions shared on the platform, which creates the landmark of believability. The study also reveals that they are sharp observers of their targets’ profiles and online activity, adapting language and timing based on this information to create a perception of legitimacy, while slowly tracking the victim's contact with outsiders. This type of social engineering takes advantage of the inherent trust that is a principle of social media. Additionally, the growth of these parasocial relationships - one-sided emotional relationships with influencers or public figures - increases users’ willingness to trust, leading to a breakdown in the ability to distinguish between real and unreal. As a result, millions of innocent users continue to fall victim to cybersecurity loopholes, and this cycle will never end until there is a dedicated investigation into the behavioral designs of common social platforms.

<https://www.researchgate.net/publication/382962280> Trust in Online Information and Psychological Well-Being

<https://www.researchgate.net/publication/385417713> Romance Scam as one of the main Challenges of cybercrime

IV. Exploiting Attention: How Social Media Design Creates Vulnerabilities (250–350 words)

- Discuss **persuasive design** (notifications, infinite scroll, likes).
- Explain how constant engagement reduces users' alertness and critical thinking.
- Cite research on **attention economy** and **habit formation** (e.g., Tristan Harris, "The Social Dilemma").

Social media companies have designed their platforms with technology to maximize distractions that engage users through persuasive design features such as notifications, endless scrolling, and likes. This engagement creates a cycle of constant familiarity which compromises users' awareness and critical thought, making it easier for users to click through malicious links or misleading information. As noted by Hayes (2014), user interface (UI) designers have put significant emphasis on creating applications that are easy, beautiful, and perceived as effortless to complete, where all of these features exist to "make users feel comfortable and to keep them engaged with the application" (Hayes, 2014, p. 3). This paradox exists in that easy to access and utilize applications can lower users' cognitive defenses, as people develop habits of clicking without caution. Hayes' research also showed that UX/UI elements, such as the "like" button and infinite scrolling, were intentionally created and optimized to ensure users felt that they were having a satisfying "experience" with an application that would encourage them to re-visit or return to the application which reinforces that the company's interests exist beyond user engagement and awareness. In addition to design, social influence compounds these experiences: users follow friends, influencers, and celebrities, while voluntarily sharing excerpts of their life, often personal data (i.e., locations, routines, relationships, opinions, etc.) to maintain social connections, gain validation, or feel a part of a trending conversation. Thus, attention economy tries to get users to be reactive not reflective. Eventually, the seamless interfaces of social media condition users to think of usability as safety. This undermines the distinction between being "convenient" and being "vulnerable." In the end, it shows how persuasive design uses human attention as a tool of manipulation—cognitively disarming the user in an environment where every tap/click/like/post has potential and roles of influencers, but often unrecognizable consequences. NEED EDIT

<https://digitalcommons.calpoly.edu/grcsp/108/>

https://www.researchgate.net/profile/Kodjo-Atiso/publication/325052793_User_Beware_Determining_Vulnerability_in_Social_Media_Platforms_for_Users_in_Ghana/links/602e685fa6fdcc37a835f03d/User-Beware-Determining-Vulnerability-in-Social-Media-Platforms-for-Users-in-Ghana.pdf

V. Social Engineering and Data Harvesting (250–350 words)

- Describe how overshared personal data fuels **phishing and identity theft**.

- Provide examples: Targeted scams, password-reset trickery, deepfake impersonations.
- Use cybersecurity reports (Verizon, FBI IC3, or academic studies on phishing trends).

By oversharing on social media, cybercriminals can obtain details needed to conduct highly targeted attacks. Users frequently share very personal information, such as birthdays, locations, employment, and relationships, without regard to it being easily harvested. As stated by Vemprala and Dietrich (2019), breach notifications and posts with technical information, threats, or indicatives related to severity spread rapidly across social networks. This phenomenon is quite strong in groups that are driven by tech-savvy use, or interest in questions of security. Cybercriminals can exploit that same social network diffusion, watching what users do and engage with as it relates to phishing campaigns, or an identity theft schemas, or even a real-world impersonation via a deepfake or fake video. For example, an attacker could use knowledge of the user's employment, coupled with knowledge of social connections, to send a password-reset email that appeared to be from the partnered workplace. Or an attacker leveraging that same social connection could create a false account in a social media group, posing as a trusted colleague. The act of personal information visibility, combined with users being accustomed to fast action card surfing and trusting prompts that more resemble their own, is faced with increased risks. Social engineering exploits psychological vulnerabilities prior to any digital action such as urgency, authority, and anticipation can even turn a reasonably cautious user into a potential victim when fraudulent actions seem to be legitimate. In addition to the previous example, the study notes that messages that amplified stress, anxiety, or threat factors had predisposed effectiveness in driving individuals to engage, which attackers could also exploit (Vemprala & Dietrich, 2019). As a consequence, cybercriminals can harvest data effectively, causing social networks themselves to become vehicles for exploitation. NEED EDIT

https://aisel.aisnet.org/hicss-52/st/cyber_threat_intelligence/8/

<https://books.google.com/books?hl=en&lr=&id=9LpawpklYogC&oi=fnd&pg=PR13&dq=Social+Engineering+and+Data+Harvesting+&ots=vdkuFRd7OS&sig=nVfwHh70cQ2UNOwfdEbsuOHK3F8>

<https://www.sciencedirect.com/science/article/abs/pii/S0306457324002875>

VI. Real-World Case Studies (250–350 words)

- Analyze specific incidents (e.g., 2020 Twitter Bitcoin scam, Cambridge Analytica).
- Explain how human behavior — not just technical flaws — played a key role.
- Tie back to the thesis: how social media designs amplified human error.

The Twitter hack is a fitting potential example for an error that has human implications using social media architecture as the container. Twitter accounts advertising a Bitcoin scam took place on July 15, 2020, between the hours of 1 p.m and 3 p.m PDT. The scam would not have occurred without the full cooperation of around 130 hyped Twitter accounts, as well as real, verified accounts of politicians, celebrities, or big companies. Scammers promised to return double to any

person who donated Bitcoin to the designated wallets for charitable purposes, and people on Twitter flocked to the account to donate their funds. In a matter of hours, and probably within a couple of minutes, about 300 deposits to one wallet surpassed \$118,000 and roughly \$61,000 was transferred from the wallet to concealed addresses. It was a social engineering attack: the hackers used phone-based spear phishing to elicit the credentials from Twitter employees, influencers, or participants that they needed to access the administration tools to reset passwords, publish tweets, and access private messages. Again, human factors came into play – employees working from home, in good faith, using their mobile phones, and believing internal communications were secure, likely contrived a low state of awareness for the hacker (Witman & Mackelprang, 2022). The design of the platform compounded the risk, with approximately 1,500 employees gaining access to powerful administrative tools, and chat-based collaboration platforms, such as Slack, providing the attackers visibility into Twitter's internal systems. The result was catastrophic: the impacted accounts had millions of followers and contributed to a loss of public trust with a single incident illustrating how social media systems can amplify consequences of individual errors. In the end, this shows us that a purely technical solution is insufficient—the security of any system is dependent on the interaction between design, organizational culture, and user behavior. NEED EDIT

<https://eric.ed.gov/?id=EJ1332789>

https://www.researchgate.net/publication/381737202_From_Tweet_to_Theft_Tracing_the_Flow_of_Stolen_Cryptocurrency

VII. Solutions and Future Directions (250–350 words)

- Discuss **education-based** solutions: cybersecurity awareness, media literacy.
- **Policy and design solutions:** ethical UX design, stronger verification systems.
- Bring in expert views (e.g., academic proposals for “human-centered cybersecurity”).
- End with a hopeful or critical statement — emphasizing awareness as the first line of defense.

VIII. Conclusion (150–200 words)

- Restate your thesis in fresh terms.
- Summarize how each section supports your main argument.
- End with a powerful line, e.g.:

“Firewalls can protect systems, but only awareness can protect people — and social media has yet to teach users how to defend themselves.”